Extra Bite #2: Practical Wire Antennas, Phased Arrays, and Ground Effects

Patterns, phased verticals, folded dipoles, OCFDs, G5RV, Zepp, and what ground and seawater do to your signal. One pass over E9C's 14 questions.

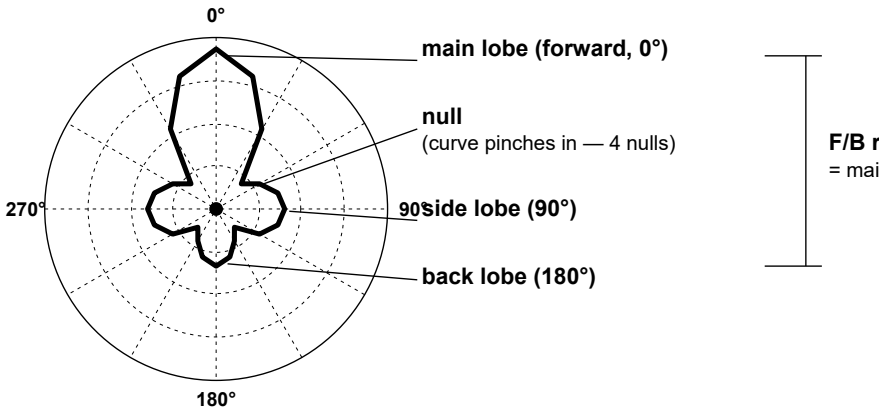
Group E9C · 14 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Why You Care

An antenna does not radiate equally in all directions. It has a **pattern**: bigger in some directions, smaller in others, sometimes zero. Knowing the pattern tells you where your signal is going. Most of E9C is about recognizing patterns from a description, naming a few classic antennas, and predicting what changes when you raise the antenna, sit it over salt water, or tilt the ground under it.

Anatomy Of A Radiation Pattern

Imagine looking down on the antenna from above. The drawing below shows a directional pattern with its parts labeled. These names appear in nearly every E9C question.



Top-down view of a generic directional antenna pattern.

Pattern Vocabulary

Word	Plain meaning
Main lobe	The biggest “bulge” in the pattern. Where most of your power goes.
Side lobes	Smaller bulges off to the sides.
Front-to-back ratio	How much louder forward is than straight backward.
Takeoff angle	Vertical angle at which the main lobe leaves the antenna. **Low** takeoff = signal heads for the horizon = better DX.
Broadside	Perpendicular (90 deg) to the line through the antenna elements.
End-fire	Along the line through the elements.
Cardioid	A heart-shaped pattern. One strong lobe forward, deep null behind.

Two Verticals, Three Patterns (E9C01-E9C03)

Take two quarter-wave verticals (the two black dots in each plot below). Change the **spacing** and the **phase** between them and the azimuth pattern changes in three predictable ways. Each plot is a top-down (azimuth) polar pattern with the array axis running east-west (90° - 270°).

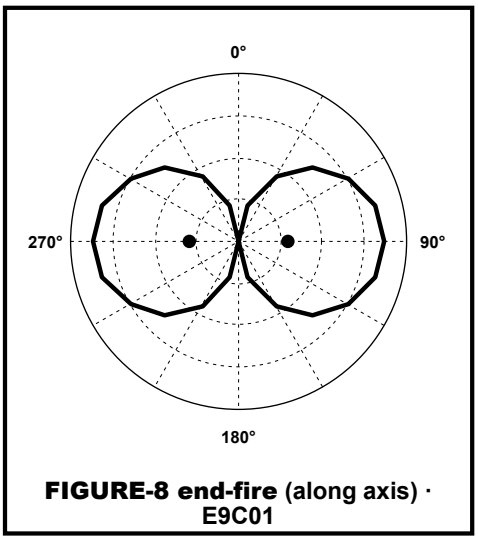
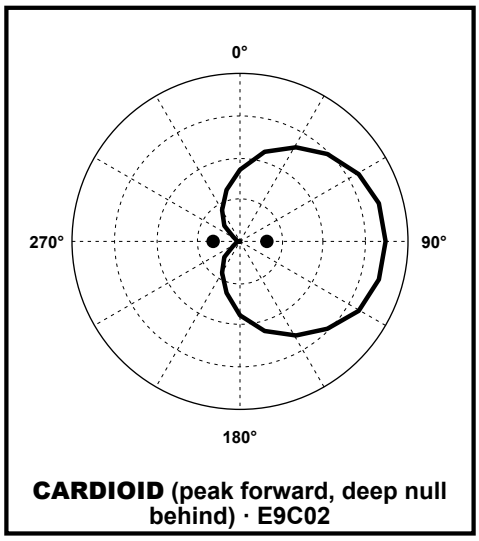


FIGURE-8 end-fire (along axis) · E9C01



CARDIOID (peak forward, deep null behind) · E9C02

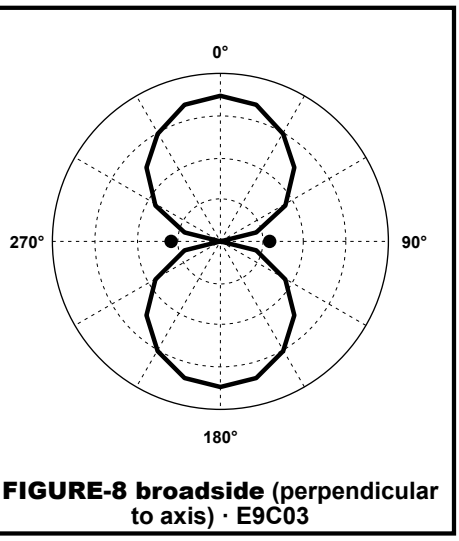
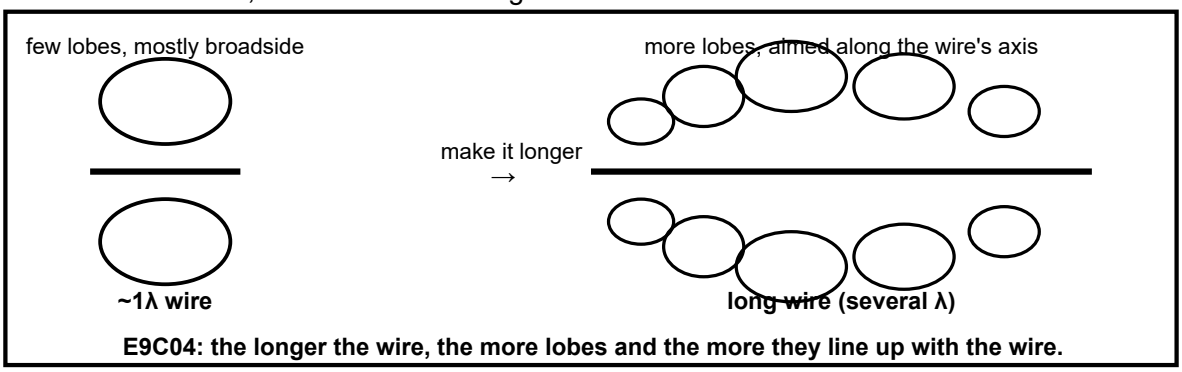


FIGURE-8 broadside (perpendicular to axis) · E9C03

Memory trick: Half-wave + 180° out → waves cancel sideways, reinforce along the line (end-fire). Half-wave + in-phase → the opposite: cancel along the line, reinforce sideways (broadside). Quarter-wave + 90° → one direction reinforces, the other cancels (heart shape).

Long-Wire And Rhombic

Make a wire antenna longer (in wavelengths) and the pattern sprouts **more lobes**, and those lobes get pulled **closer to the axis of the wire**, not broadside. A long wire fires off its ends.



E9C04: the longer the wire, the more lobes and the more they line up with the wire.

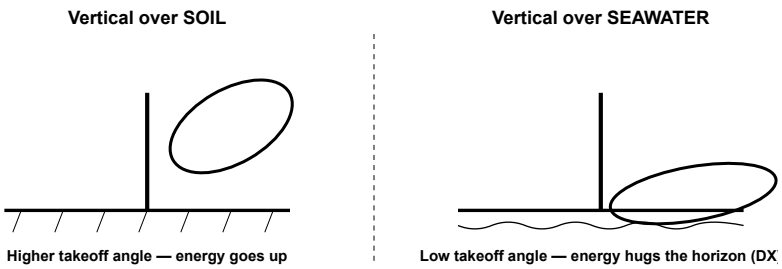
Terminating resistor at the far end of a rhombic or long wire kills the reflected wave that would have come back. The pattern goes from **bidirectional to unidirectional**. Trade: some power burned in the resistor. (E9C06)

Named Antennas You Need To Recognize

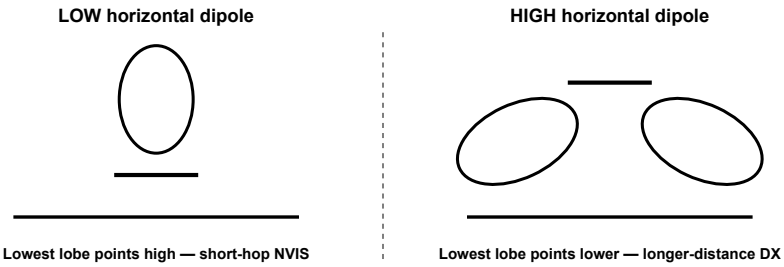
Antenna	What it is	Q
Folded dipole	A half-wave dipole with an extra parallel wire connecting the two ends. The fold raises the feed impedance to about **300 ohms**, which is why folded dipoles match 300-ohm twin-lead.	E9C07, E9C08
Off-center-fed dipole (OCFD)	A dipole fed somewhere between the center and one end. The off-center feed lands at a similar impedance on several harmonically related bands, so one OCFD can work multi-band.	E9C05
G5RV	A center-fed wire about 102 ft long, fed through a specific length of open-wire line into a balun, then coax. A classic multi-band wire.	E9C09
Zepp	An end-fed half-wavelength dipole. (Name comes from the trailing wire antennas on old Zeppelins.)	E9C10
Extended double Zepp	A center-fed wire that is **1.25 wavelengths** total. The extra length squeezes the main lobe narrower for some forward gain over a plain half-wave dipole.	E9C12

How The Ground And Sky Change The Pattern

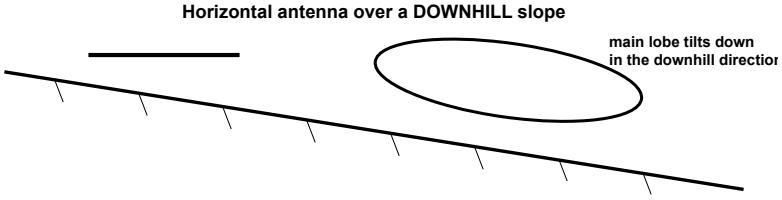
These are **side-view** pictures: ground is at the bottom, the sky is above, and the antenna is somewhere in between. The question is at what angle the energy leaves the antenna.



E9C11: Salt water is a near-perfect mirror; soil is a poor one. Over seawater, low-angle radiation increases.



E9C13: Raise a horizontal antenna and the lowest-lobe takeoff angle decreases.



E9C14: A sloping ground tilts the effective ground plane; the takeoff angle decreases on the downhill side. Hilltop stations love this.

Mental Cheat Sheet

- Two verticals 1/2 wave apart, 180° out → figure-8 end-fire.
- Two verticals 1/4 wave apart, 90° out → cardioid.
- Two verticals 1/2 wave apart, in phase → figure-8 broadside.
- Long wire gets longer → more lobes, aligned with the wire.
- Terminated long wire / rhombic → bidirectional becomes unidirectional.
- Folded dipole → 300 ohms feed; OCFD → similar Z on multiple bands.
- G5RV → center-fed wire + open-wire matching + balun + coax.
- Zepp → end-fed half-wavelength; Extended double Zepp → center-fed 1.25λ.
- Vertical over seawater vs soil → more low-angle radiation.
- Horizontal higher → lower takeoff angle.
- Horizontal over downhill slope → takeoff angle drops downhill.

Common Traps

- The two-vertical questions look almost identical. Read **spacing** and **phase** carefully, then look up the picture.
- “Broadside” and “end-fire” are opposites and the test offers both. Broadside = perpendicular to the line of elements. End-fire = along the line of elements.
- The terminating resistor on a rhombic does not change polarization or feed-line standing wave; it changes the **pattern from bidirectional to unidirectional**.
- Folded dipoles are still half-wave antennas; the fold only raises the feed impedance.
- Raising a horizontal antenna **lowers** the takeoff angle. Students sometimes invert this.
- Over seawater, low-angle radiation **increases**, not decreases.

Takeaway: An antenna’s pattern depends on its geometry, its phasing, and what is under it. Learn the three two-vertical pictures, recognize the named wire antennas, and remember that both height and seawater push the energy lower toward the horizon.

Bite 2 Question Bank (E9C)

14 questions. Read, decide, check the answer key at the end.

E9C01. What type of radiation pattern is created by two 1/4-wavelength vertical antennas spaced 1/2-wavelength apart and fed 180 degrees out of phase?
A. Cardioid.
B. Omni-directional.
C. A figure-eight broadside to the axis of the array.
D. A figure-eight oriented along the axis of the array.

E9C02. What type of radiation pattern is created by two 1/4-wavelength vertical antennas spaced 1/4-wavelength apart and fed 90 degrees out of phase?
A. Cardioid.
B. A figure-eight end-fire along the axis of the array.
C. A figure-eight broadside to the axis of the array.
D. Omni-directional.

E9C03. What type of radiation pattern is created by two 1/4-wavelength vertical antennas spaced 1/2-wavelength apart and fed in phase?
A. Omni-directional.
B. Cardioid.
C. A figure-eight broadside to the axis of the array.
D. A figure-eight end-fire along the axis of the array.

E9C04. What happens to the radiation pattern of an unterminated long wire antenna as the wire length is increased?
A. Fewer lobes form with the major lobes increasing closer to broadside to the wire.
B. Additional lobes form with major lobes increasingly aligned with the axis of the antenna.
C. The elevation angle increases, and the front-to-rear ratio decreases.
D. The elevation angle increases, while the front-to-rear ratio is unaffected.

E9C05. What is the purpose of feeding an off-center-fed dipole (OCFD) between the center and one end instead of at the midpoint?
A. To create a similar feed point impedance on multiple bands.
B. To suppress off-center lobes at higher frequencies.
C. To resonate the antenna across a wider range of frequencies.
D. To reduce common-mode current coupling on the feed line shield.

E9C06. What is the effect of adding a terminating resistor to a rhombic or long-wire antenna?
A. It reflects the standing waves on the antenna elements back to the transmitter.
B. It changes the radiation pattern from bidirectional to unidirectional.
C. It changes the radiation pattern from horizontal to vertical polarization.
D. It decreases the ground loss.

E9C07. What is the approximate feed point impedance at the center of a two-wire half-wave folded dipole antenna?
A. 300 ohms.
B. 72 ohms.
C. 50 ohms.
D. 450 ohms.

E9C08. What is a folded dipole antenna?
A. A dipole one-quarter wavelength long.
B. A center-fed dipole with the ends folded down 90 degrees at the midpoint of each side.
C. A half-wave dipole with an additional parallel wire connecting its two ends.
D. A dipole configured to provide forward gain.

E9C09. Which of the following describes a G5RV antenna?
A. A wire antenna center-fed through a specific length of open-wire line connected to a balun and coaxial feed line.
B. A multi-band trap antenna.
C. A phased array antenna consisting of multiple loops.
D. A wide band dipole using shorted coaxial cable for the radiating elements and fed with a 4:1 balun.

E9C10. Which of the following describes a Zepp antenna?
A. A horizontal array capable of quickly changing the direction of maximum radiation by changing phasing lines.
B. An end-fed half-wavelength dipole.
C. An omni-directional antenna commonly used for satellite communications.
D. A vertical array capable of quickly changing the direction of maximum radiation by changing phasing lines.

E9C11. How is the far-field elevation pattern of a vertically polarized antenna affected by being mounted over seawater versus soil?
A. Radiation at low angles decreases.
B. Additional lobes appear at higher elevation angles.
C. Separate elevation lobes will combine into a single lobe.
D. Radiation at low angles increases.

E9C12. Which of the following describes an extended double Zepp antenna?
A. An end-fed full-wave dipole antenna.
B. A center-fed 1.5-wavelength dipole antenna.
C. A center-fed 1.25-wavelength dipole antenna.
D. An end-fed 2-wavelength dipole antenna.

E9C13. How does the radiation pattern of a horizontally polarized antenna vary with increasing height above ground?
A. The takeoff angle of the lowest elevation lobe increases.
B. The takeoff angle of the lowest elevation lobe decreases.
C. The horizontal beamwidth increases.
D. The horizontal beamwidth decreases.

E9C14. How does the radiation pattern of a horizontally-polarized antenna mounted above a long slope compare with the same antenna mounted above flat ground?
A. The main lobe takeoff angle increases in the downhill direction.
B. The main lobe takeoff angle decreases in the downhill direction.
C. The horizontal beamwidth decreases in the downhill direction.
D. The horizontal beamwidth increases in the uphill direction.

Answer Key

E9C01: D	E9C02: A	E9C03: C
E9C04: B	E9C05: A	E9C06: B
E9C07: A	E9C08: C	E9C09: A
E9C10: B	E9C11: D	E9C12: C
E9C13: B	E9C14: B	